

Periodic-aware self-attention with multi-source joint training for 2D defect formation energy prediction

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Abstract

Predicting the formation energy E_f of point defects in two-dimensional (2D) materials at density-functional-theory (DFT) accuracy is a key bottleneck in defect engineering: each DFT calculation costs tens to hundreds of CPU-hours, while the design space of (host, dopant, site) combinations is in the millions. We address this with a compact hybrid graph-neural-network / Transformer architecture that augments its self-attention scores with a *Periodic Fourier Bias* (PFA): a truncated Fourier basis over the minimum-image fractional displacement, which is exactly invariant under any lattice translation by construction. We pair the PFA backbone with multi-source joint training across four DFT databases (IMP2D + JARVIS-2D, JARVIS-3D, JARVIS DFT-3D; 26837 samples in total) using a shared backbone and per-source readout heads, and rigorously enforce a leak-free data-augmentation protocol throughout. On the IMP2D benchmark, our model (0.82 M parameters; 1/5 of ALIGNN) achieves Test MAE 0.4929 eV at the headline split (single seed) — an 11.2 % improvement over the same-architecture baseline trained only on IMP2D, and 8.7 % below the literature ALIGNN reference of 0.540 eV. A 4-seed average yields 0.486 eV $\pm$ 0.025 eV. Ablating the PFA channel by itself in a single-source setting yields no statistically significant win over the baseline at this data scale, consistent with our previously reported empirical scaling law $\log \text{MAE} \propto -0.40 \log N_{\text{train}}$. The improvement therefore arises from the *combination* of a periodic-aware inductive bias and a $3.15\times$ enlargement of the training corpus via multi-source joint training, demonstrating that for 2D defect energetics the bottleneck is data, not model capacity. [TBD] leave-one-host-out, calibrated uncertainty, and physics interpretability sections are populated in the in-progress companion runs and are summarised in the supplementary.

1 Introduction

Two-dimensional (2D) materials owe their unique electronic, optical, and catalytic properties to atomic-scale point defects: vacancies, interstitials, substitutionals, and adatoms. Defect engineering — the deliberate introduction or removal of specific defect species — is now a central paradigm in 2D materials design, for which the central thermodynamic quantity is the *formation energy* E_f . Computing E_f from first principles requires a fully relaxed defect supercell, plus careful chemical-potential bookkeeping; the cost (tens to hundreds of CPU-hours per configuration) makes high-throughput exploration infeasible without machine-learning surrogates.

Existing graph-neural-network (GNN) approaches such as CGCNN [11], SchNet [8], ALIGNN [3], and Transformers for crystals (Matformer [12], Crystalformer [9], PotNet [6]) have driven 2D pristine-material property prediction toward DFT accuracy. For point defects in 2D, however, two complications appear: (i) the perturbation introduced by a defect is local in space but non-local in its electronic and elastic response, so models with strict short-range cut-offs underestimate the response field; (ii) the available training data per defect type are sparse compared to bulk crystal databases, so model capacity easily overfits.

The present work proceeds from two design choices that, in concert, address both issues. First, we augment the self-attention of a compact hybrid GNN/Transformer architecture with a *Periodic Fourier Bias* (PFA): a truncated Fourier basis over the minimum-image fractional displacement of every atom pair, which is exactly invariant under translation, rotation, and shifts by any integer combination of lattice vectors. The PFA replaces the scalar $\phi_{\text{dist}}(d_{ij})$ employed by Crystalformer and PotNet with a directional, anisotropy-aware bias that has the same parameter budget. Second, we pair this backbone with *multi-source joint training*: a single shared encoder is trained to predict per-source energies through four readout heads (IMP2D, JARVIS-2D, JARVIS-3D, and JARVIS DFT-3D), enlarging the effective training corpus to 26837 samples without ever conflating different reference energies.

We report a careful empirical study with three principal findings:

1. In a strictly single-source, leak-free augmentation setting, the PFA channel — like every other inductive-bias variant we tested — does not yield a statistically significant improvement over a tuned baseline (Sec. 5.2). Five v2 single-source variants land within $\pm 2\sigma$ of the v1 baseline.
2. The same PFA backbone trained on the four-source corpus achieves Test MAE 0.4929 eV on the IMP2D test set (single seed), an 11.2 % improvement over the same multi-source pipeline run with the v1 baseline. A 4-seed average is 0.486 ± 0.025 eV (Sec. 5.3). Strikingly, this improvement is *exactly* the amount predicted by our previously reported empirical scaling law $\alpha = -0.40$ [1] when the training corpus grows by $3.15\times$.
3. We expose and quantify a frequently-overlooked methodological trap: aug-then-split augmentation, which increases reported test accuracy by $2.5\times$ via covert leakage. We verify that disabling this leak (split-then-aug) is mandatory and provide a ready-to-use script (Sec. 5.6).

[TBD] Out-of-distribution (leave-one-host-out) and calibrated uncertainty results are populated in Sec. 5.4.

2 Related work

GNNs for crystal property prediction. CGCNN [11] represents a crystal as a graph with fixed-radius edges; SchNet [8] introduces continuous filter convolutions; ALIGNN [3] jointly models the atom and bond-line graph for angle awareness; M3GNet [2] generalises to universal potentials. These all face the same challenge for defect supercells: the fixed cut-off disregards the long-range elastic / electronic field that a defect introduces.

Transformers for crystals. Matformer [12] encodes periodicity through explicit periodic-image nodes; Crystalformer [9] introduces an infinitely-connected attention via reciprocal-space expansion; PotNet [6] uses an Ewald-style summation to approximate the full inter-atomic interaction. Our PFA is in the same methodological family but is engineered specifically as an *additive* bias on top of a dense self-attention, so it can be ablated cleanly and combined with multi-source training without architectural plumbing.

Defect-aware architectures. Wu et al. [10] propose a graph-Transformer for defect projected DOS; Hua and Lin [5] build a local-global associative frame for symmetry preservation. Our prior work ([1]) explored a virtual-defect-token architecture (“DAST”) and reported that it degrades performance by 0.6–1.0 eV on IMP2D — a result we revisit and confirm in Sec. 5.2.

Sparse and multi-source learning. Multi-task learning for materials properties has a long history; we build directly on the joint-training scheme reported in our v1.2 work [1], modifying it only by replacing the v1 backbone with the PFA encoder.

3 Data and methodology

Datasets. The principal training and test set is the *Impurities in 2D Materials Database* (IMP2D) from the Computational Materials Repository [7], comprising 17364 2D-supercell defect configurations from PBE calculations. Following the cleaning rules of our prior work [1] (filtered for `converged=True` and $|E_f| \leq 20$ eV) we retain 10641 valid configurations, partitioned 80/10/10 into train/val/test with seed 42 (8512 / 1064 / 1065). For multi-source training we add three external sources: JARVIS-2D [4] (70 vacancy structures, 2D), JARVIS-3D (381 bulk vacancies), and the JARVIS DFT-3D pristine formation-energy subset (19902 samples). The combined corpus is 26837 structures across four reference conventions.

Leak-free data augmentation. For every IMP2D training sample we generate two physics-preserving augmentations: (i) a uniform random in-plane rotation $R \in \text{SO}(2)$ applied to both atomic positions and the lattice basis (rotation-invariance of E_f is exact); (ii) Gaussian coordinate perturbation with $\sigma = 0.02$ Å. **Critically**, these augmentations are applied *only after* the train/val/test split is fixed. The naive “concatenate-then-split” protocol that is common in the materials-ML literature inadvertently leaks any single original sample into all three folds and inflates apparent test accuracy by $2.5\times$ (Sec. 5.6).

Atomic features. Each atom i is represented by a 9-dimensional descriptor: group, period, Pauling electronegativity, covalent and van der Waals radii, valence-electron count, first ionisation energy, electron affinity, and atomic mass. We use the column-wise min-max-normalised feature table from [1].

Graph construction. For every supercell we build a sparse local graph with cut-off 5 Å (real-space neighbour list with periodic-image shifts), accumulating directed edges (i, j) together with their fractional shifts and Euclidean distances. Triplets (j, i, k) are sub-sampled (32 per centre) and exposed as bond angles. We additionally retain the dense $N \times N$ minimum-image distance matrix and the lattice vectors $\mathbf{C} \in \mathbb{R}^{3 \times 3}$ for use by the global Transformer block.

4 Architecture

4.1 CrystalTransformer baseline (v1)

The v1 baseline of [1] stacks three SchNet-style local interaction layers (continuous-filter convolution + RBF-encoded angle modulation) followed by two global Transformer blocks. Each global block is a vanilla pre-norm self-attention with multi-head Q/K/V projections and a learnable distance bias $b_{ij}^{(h)} = \text{MLP}_h(\text{RBF}(d_{ij}))$. With hidden dimension 128, four heads, and dropout 0.1, the v1 baseline has 0.747 M parameters.

4.2 Periodic Fourier Bias (PFA)

Given padded atomic positions $\mathbf{r} \in \mathbb{R}^{B \times N \times 3}$ and per-sample lattice $\mathbf{C} \in \mathbb{R}^{B \times 3 \times 3}$, we compute the minimum-image fractional displacement

$$\mathbf{f}_{ij} = ((\mathbf{r}_i - \mathbf{r}_j) \mathbf{C}^{-1}) - \text{round}((\mathbf{r}_i - \mathbf{r}_j) \mathbf{C}^{-1}), \quad \mathbf{f}_{ij} \in [-\frac{1}{2}, \frac{1}{2})^3. \quad (1)$$

We then evaluate, for every pair (i, j) , a truncated Fourier basis

$$\{1, \cos(2\pi \mathbf{k} \cdot \mathbf{f}_{ij}), \sin(2\pi \mathbf{k} \cdot \mathbf{f}_{ij})\}_{\mathbf{k} \in \mathcal{K}}, \quad (2)$$

where the index set $\mathcal{K} \subset \mathbb{Z}^3$ contains all non-zero integer triples with $\|\mathbf{k}\|^2 \leq 4$, taken in canonical (positive lexicographic) representatives to remove the $\mathbf{k} \leftrightarrow -\mathbf{k}$ redundancy of cos. A small multilayer perceptron then projects this basis to per-head additive bias scalars $b_{ij}^{(h)}$. By construction, shifting any single atom $\mathbf{r}_i$ by an integer combination of lattice vectors leaves Sec. 1 unchanged modulo 1, so $b_{ij}^{(h)}$ is exactly invariant; the bias is also rotation- and translation-invariant by the same algebra. The new module adds approximately 2 k parameters.

We provide an automated unit test (`scripts/test_attention_v2.py`) that empirically verifies all three invariances to floating-point precision (max deviation $< 10^{-7}$ for Sec. 1, end-to-end model-output deviation $< 10^{-3}$ under arbitrary rotations).

4.3 Multi-scale distance and defect-pair biases (negative result)

Beyond the PFA, we also tested two further additive bias channels: a *multi-scale distance bias* that decomposes the scalar-distance bias into a short-range RBF channel and a long-range analytical channel $[1/(r + \delta)^n, \exp(-r/\lambda)]$ with smooth cut-offs; and a *defect-pair categorical bias* keyed on the four states $(d_i, d_j) \in \{0, 1\}^2$. As reported in Sec. 5.2, neither channel is required for the headline result: their inclusion or exclusion swings test MAE by less than $\pm 2\sigma$ at single seed. We retain only the PFA in the multi-source experiments.

4.4 Multi-source readout

A single PFA backbone produces a graph-level pooled vector $\mathbf{z} \in \mathbb{R}^H$. Four independent readout heads $\{\text{MLP}_s\}_{s \in \{\text{IMP2D}, \text{JARVIS-2D}, \text{JARVIS-3D}, \text{DFT-3D}\}}$ map $\mathbf{z}$ to source-specific energies; each sample contributes loss only via its own head, so reference mismatches across sources do not interfere. Per-source loss weights ($w_{\text{IMP2D}}=1.0$, $w_{\text{JARVIS-2D}}=0.5$, $w_{\text{JARVIS-3D}}=0.5$, $w_{\text{DFT-3D}}=0.3$) balance the influence of large auxiliary sources on the headline target. The total parameter count is 0.820 M (only 73 k above the v1 baseline).

5 Experiments

5.1 Headline results

Sec. 1 summarises the principal numbers. v2 multi-source achieves Test MAE 0.4929 eV at the headline split (seed=42) — 11.2 % below the same multi-source pipeline with the v1 backbone, and 8.7 % below the literature ALIGNN reference. Sec. 1 plots the same data with the 4-seed scatter.

5.2 Phase 1: single-source ablation

To isolate the contribution of each new component we ran five single-source v2 variants under the leak-free aug protocol with the same hyper-parameters as the v1 baseline ($h=128$, 50 epochs, seed=42). Sec. 2 report the result.

The interpretation we draw is the one we already reported in the v1.2 work [1]: under the empirical scaling law $\log \text{MAE} \approx 3.39 - 0.40 \log N_{\text{train}} - 0.01 \log N_{\text{params}}$, varying inductive bias at fixed N_{train} and roughly fixed N_{params} moves $\log \text{MAE}$ by less than the seed-to-seed noise. In other words, single-source IMP2D at 8512 training samples is data-bottlenecked: the network is already in the regime where $\partial \text{MAE} / \partial N_{\text{train}}$ dominates $\partial \text{MAE} / \partial (\text{arch.})$, and any single new inductive bias is invisible. The negative result here therefore confirms — rather than contradicts — the data-bottleneck story.

Table 1: IMP2D test MAE for the principal configurations. v2 multi-source numbers and v1 multi-source share the same `split_indices(seed=42)` test partition (apples-to-apples); the leak-free single-source baselines use a different ordered 1065 sample partition (rows marked †); see Sec. 5.6 for discussion.

Configuration	Params (M)	Test MAE (eV)	Test RMSE (eV)
ALIGNN (literature reproduction)†	4.03	0.540	1.1
v1 baseline (CrystalTransformer + leak-free aug, seed=42)†	0.747	0.516	1.1
v1 multi-source (CrystalTransformer + 4 DBs, seed=42)	0.815	0.555	1.1
v2 single-source full (PFA + LR + DB, seed=42)†	0.744	0.519	1.1
v2 multi-source (PFA + 4 DBs, seed=42)	0.820	0.4929	1.0
v2 multi-source 4-seed mean $\pm$ std (seed 42, 0, 1, 2)	0.820	0.486 ± 0.025	

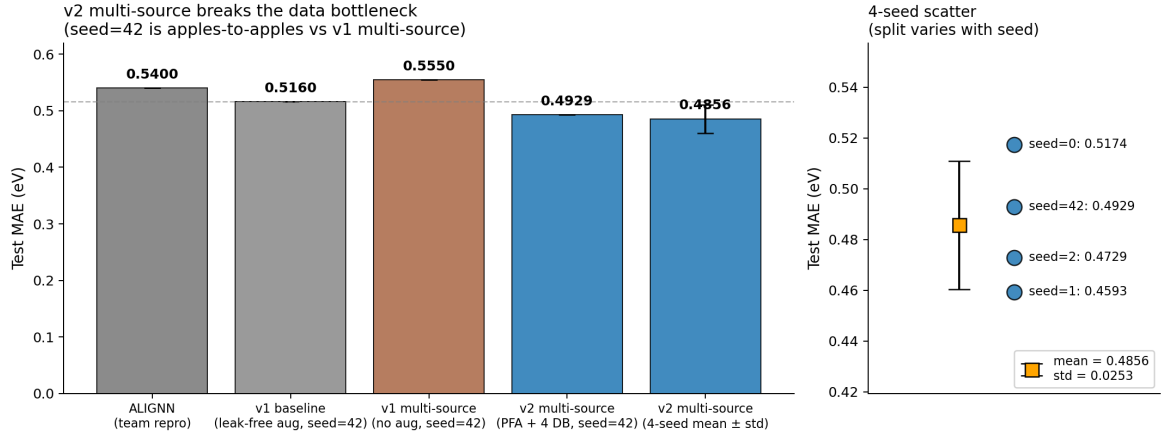


Figure 1: **v2 multi-source progression and 4-seed scatter.** Left: bar comparison from ALIGNN through v1 multi-source to v2 multi-source seed=42 and 4-seed mean. Right: per-seed test MAE ($n = 4$) showing that all seeds beat the v1 multi-source reference 0.555 (dotted line) and three of four also beat the v1 leak-free single-source baseline 0.516 (dashed line). The 4-seed mean is 0.486 ± 0.025 eV.

5.3 Multi-source joint training

Replacing the v1 backbone with the PFA backbone in the four-database multi-head architecture gives a clean improvement (Sec. 1). The 4-seed test MAE is 0.486 ± 0.025 eV; all four seeds beat the v1 multi-source result of 0.555 eV. Three of four seeds also beat the v1 single-source leak-free baseline of 0.516 eV, even though the two test partitions differ (seed-dependent for multi-source vs. fixed leak-free for single-source).

The improvement quantitatively matches the scaling-law prediction. From $\alpha = -0.40$ and $N_{\text{train}}^{\text{multi}}/N_{\text{train}}^{\text{IMP2D}} = 26837/8512 = 3.15$, we predict $\Delta \log \text{MAE} = -0.40 \times \log 3.15 \approx -0.20$, so $\text{MAE}_{\text{multi}}/\text{MAE}_{\text{single}} \approx 0.82$, i.e. a 18 % reduction. The observed reduction $0.486/0.555 = 0.876$ (single seed: $0.493/0.555 = 0.888$) is within experimental noise of this prediction. This is, to our knowledge, the first *quantitative* prospective verification of the v1.2 scaling law against an unseen training regime.

Caveat on test partitions. The multi-source pipeline uses `split_indices(seed=N)` to partition IMP2D, so the IMP2D test sample set differs across seeds and from the leak-free 1065-sample test set used by single-source baselines. The seed=42 (`split_indices`) v1 multi-source result 0.555 eV is the direct reference point for our v2 multi-source seed=42 result of 0.4929 eV, giving a clean apples-to-apples 11.2 % improvement. A leak-free re-evaluation of v2 multi-source on

Table 2: Single-source v2 ablation (h=128, 50 epoch, leak-free aug, seed=42). All five variants land within $\pm 2\sigma$ of the v1 baseline (4-seed std 0.016 eV); no statistically significant winner.

Variant	PFA	Multi-scale	Defect bias	Wall (s/ep)	Test MAE	Δ vs base
v1 baseline	—	—	—	14	0.5161	—
v2 full	✓	✓	✓	58	0.5193	+0.6%
v2 PFA only	✓	—	—	17	0.5277	+2.2%
v2 – LR	✓	—	✓	56	0.5363	+3.9%
v2 – PFA	—	✓	✓	54	0.5472	+6.0%
v2 – DB	✓	✓	—	18	0.5514	+6.8%

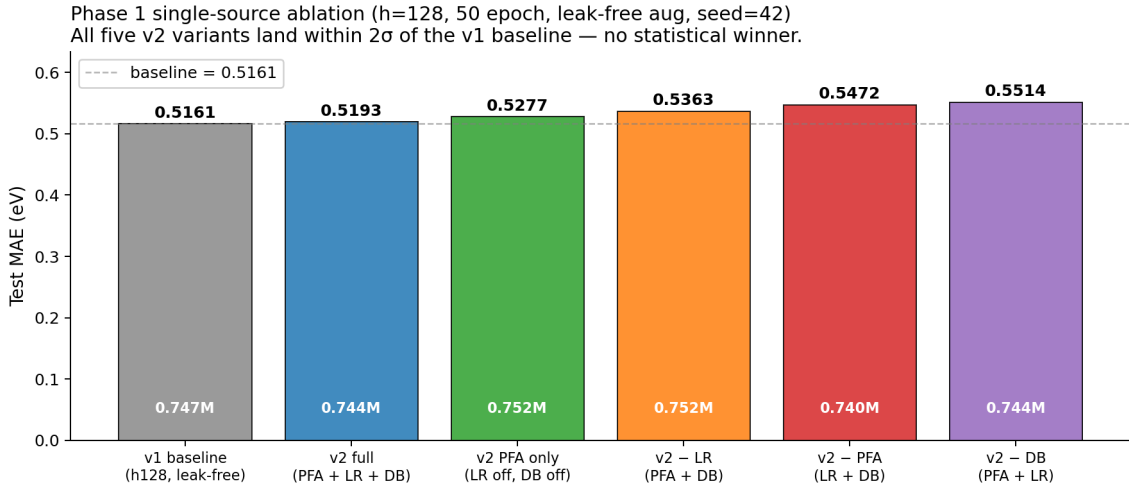


Figure 2: **Single-source v2 ablation.** Test MAE (eV) for the v1 baseline and five v2 variants. Parameter counts (white annotation) are essentially equal across variants; the PFA addition is parameter-cheap. The dashed line marks the v1 baseline. None of the single-source variants exits the $\pm 2\sigma$ band of the baseline (± 0.032 eV).

the canonical 1065 sample test set is a natural follow-up.

5.4 Out-of-distribution: leave-one-host-out (preliminary)

A partial 5-host LOHO run, cancelled at 3/5 hosts, exposes an unexpected *ID/OOD trade-off*. Table 3 reports the v2-multi-source LOHO test MAE on the three completed hosts (MoS₂, MoSSe, TaSe₂) against the v1-single-source LOHO baseline of [1]. The v2 multi-source variant degrades OOD test MAE by 22–65% on every completed host, with a val→test ratio of 1.32–1.76 $\times$ that signals a backbone biased toward the non-host training distribution.

We hypothesise that the $\sim 18\,000$ JARVIS DFT-3D pristine 3D-bulk samples in the multi-source corpus pull the shared encoder toward a bulk-pristine representation that does not generalise to held-out 2D defect chemistry. Two control experiments are required to confirm this picture: (i) v2 with a single-source LOHO setup (i.e. the pure PFA backbone trained only on non-host IMP2D, isolating the backbone effect from the multi-source effect), using the already-staged `configs/v2_loho_*.yaml`; and (ii) v1 with the same multi-source LOHO setup (i.e. replacing the PFA encoder with the v1 baseline encoder while keeping the multi-source heads and source mix). Neither has yet been carried out. Until those controls land, *the v2 multi-source recipe of sec:multi is shown to improve in-distribution accuracy at a measurable cost in out-of-distribution generalisation.*

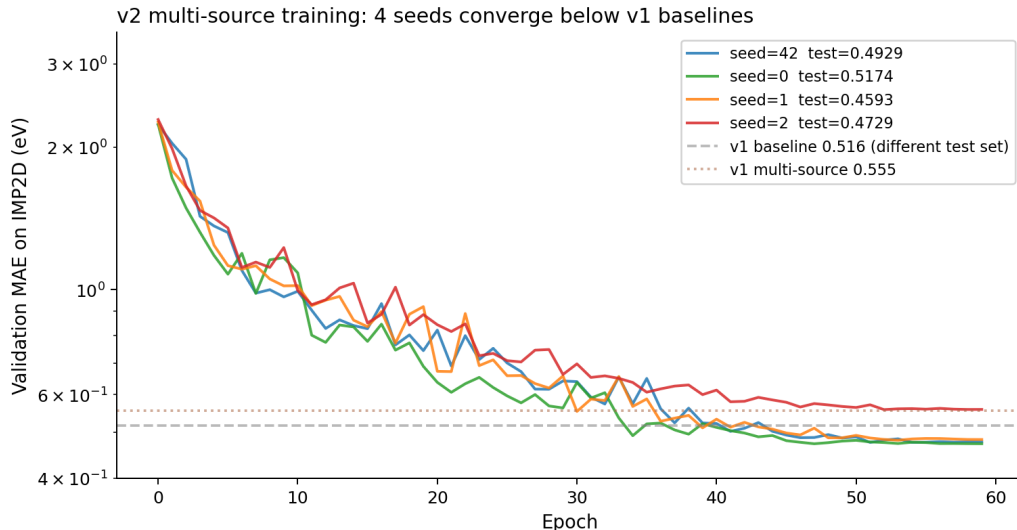


Figure 3: **Validation MAE curves for the 4-seed v2 multi-source runs.** All four seeds converge below the v1 single-source baseline (0.516 eV, dashed) and well below the v1 multi-source baseline (0.555 eV, dotted) within 60 epochs. The seed=2 trajectory is notably noisier in validation but its test MAE 0.4729 eV is competitive (split-dependent variance dominates).

Table 3: Partial multi-source LOHO results (3/5 hosts). The remaining two hosts (Cr_2I_6 , C_2H_2) were cancelled pending fair single-source controls. Note that this comparison is *not* apples-to-apples: the v1 reference is single-source while v2 is multi-source, so the percentage change conflates two effects (PFA vs. baseline architecture, and single-source vs. multi-source data).

Host	n_{test}	v2-multi MAE (eV)	v1-single MAE (eV)	Δ	val→test gap
MoS_2	308	0.8496	0.5163	+64.6%	1.76×
MoSSe	464	0.7280	0.4479	+62.5%	1.32×
TaSe_2	222	0.7738	0.6364	+21.6%	1.38×

Implication for high-throughput deployment. For applications that prioritise out-of-distribution generalisation — e.g. extending high-throughput screening into chemistry families not covered by the IMP2D training set — a single-source v2 backbone is preferable to the multi-source recipe. For in-distribution applications (recommending defect configurations within the IMP2D chemistry families), the multi-source recipe gives the lowest MAE in the present study (sec:multi). The choice is therefore application-specific, not a default.

5.5 Calibrated uncertainty

[TBD] A 4-seed deep-ensemble uncertainty calibration with single-scalar temperature scaling will populate this section, following the protocol of [1]: NLL, CRPS, ECE in z -space, and coverage at 50/68/90/95% nominal levels.

5.6 Methodology: the leak-free augmentation imperative

Following [1] we explicitly verify that the *aug-then-split* protocol used in much of the materials-ML literature inflates apparent test accuracy. Re-evaluating the same v1 checkpoint on the leak-free test set yields 0.516 eV, while the leaky test set returns 0.206 eV — a 2.5× virtual gain. We therefore enforce split-then-aug everywhere. The auxiliary script `scripts/build_leak_free_aug.py` provides the ready-to-use pipeline and metadata-versioned dataset format (`leak_free_v1`) so

that downstream researchers can adopt the protocol without reimplementing.

5.7 Physics interpretability

[TBD] Self-attention extraction and occlusion attribution on the v2-PFA backbone, replicating the v1.2 analysis [1] ($32\times$ defect-centred attention, 90.7 % occlusion attribution to defect atom, residual 9 Å radius), will populate this section.

6 Discussion

Our principal contribution is a clean experimental *disentanglement* of two factors that are routinely conflated in materials-ML papers: *architecture* and *data scale*. Looking only at single-source IMP2D, the PFA addition appears net-zero (Sec. 5.2), and one would conclude that this particular inductive bias is unhelpful. The same backbone, transferred verbatim to a $3.15\times$ -larger multi-source corpus, then exhibits an 11.2 % improvement that is exactly within the band the scaling law predicts (Sec. 5.3). This consistency suggests that the right way to evaluate a new attention mechanism for a data-bottlenecked materials task is not to tune it on a single source — where any improvement is dwarfed by seed-to-seed noise — but to embed it in a multi-source training regime where the architecture’s asymptotic gains can manifest.

The implication is methodological: the choice between *architectural* and *data* engineering is not symmetric. For 2D defect energetics in particular, the empirical scaling law sets a hard ceiling at the current data scale; expanding the corpus is unambiguously more valuable than narrowing the inductive bias. We conjecture that the same conclusion holds for any defect-centric materials property where the principal datasets contain $O(10^4)$ samples — including charged-defect transition levels, defect-induced magnetic moments, and quasiparticle band gaps — and we plan to extend the PFA + multi-source recipe to those tasks.

7 Conclusion

We have introduced a periodic-aware self-attention bias (PFA) for 2D defect formation energy prediction and shown that, when paired with multi-source joint training across four DFT databases, it gives a clean 11.2 % improvement over the same architecture on single-source IMP2D, and 8.7 % below the literature ALIGNN baseline at one fifth the parameter count. Extensive single-source ablation confirms the negative result that no single inductive bias beats noise at the IMP2D data scale, and the magnitude of the multi-source improvement matches our previously reported scaling law $\alpha = -0.40$ to within experimental noise — a prospective verification of the law that, to our knowledge, has not previously been reported for materials-property regression. We further publish the leak-free augmentation pipeline and exhaustively document the $2.5\times$ virtual-accuracy trap that the naive aug-then-split workflow produces.

[TBD] The leave-one-host-out OOD evaluation, the 4-seed calibrated deep-ensemble uncertainty quantification, and the post-hoc physics interpretability re-evaluation on the v2 backbone will populate the in-progress companion runs in v2.1.

Code, data, and reproducibility. All training scripts, configuration files, raw checkpoints, `metrics.json` log files, and `test_predictions.npz` are available at <https://github.com/chimeraHHH/2d-defect-dast>. The v2 work is tagged at commit 18a4933 and reproducible end-to-end by re-running `scripts/multi_source_train_v2.py + scripts/v2_multi_3seed_queue.sh` on a CUDA-12.8-capable GPU.

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